

From English to Foreign Languages: Transferring Pretrained Language Models

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Abstract

Pretrained models have demonstrated their effectiveness in many downstream natural language processing (NLP) tasks. The availability of multilingual pretrained models enables zero-shot transfer of NLP tasks from high resource languages to low resource ones. However, recent research in improving pretrained models focuses heavily on English. While it is possible to train the latest neural architectures for other languages from scratch, it is *undesirable* due to the required amount of compute. In this work, we tackle the problem of transferring an existing pretrained model from English to other languages under a *limited computational budget*. With a single GPU, our approach can obtain a foreign BERT_{BASE} model within a day (20 hours) and a foreign BERT_{LARGE} within two days (46 hours). Furthermore, evaluating our models on six languages, we demonstrate that our models are better than multilingual BERT on two zero-shot tasks: natural language inference and dependency parsing. Our code is available at <https://github.com/anonymized>.

1 Introduction

Pretrained models (Devlin et al., 2019; Peters et al., 2018) have received much of attention recently thanks to their impressive results in many downstream NLP tasks. Additionally, multilingual pretrained models enable many NLP applications for other languages via zero-shot cross-lingual transfer. Zero-shot cross-lingual transfer has shown promising results for rapidly building applications for low resource languages. Wu and Dredze (2019) show the potential of multilingual BERT (Devlin et al., 2019) in zero-shot transfer for a large number of languages from different language families on five NLP tasks, namely, natural language inference, document classification, named entity recognition, part-of-speech tagging, and dependency parsing.

Although multilingual models are an important piece for building up language technology in many languages, recent research on improving pretrained models puts much emphasis on English (Radford et al., 2019; Dai et al., 2019; Yang et al., 2019). The current state of affairs makes it difficult to translate developments in pre-training from English to non-English languages. To our best knowledge, there are only three publicly available multilingual pretrained models to date:

1. The multilingual BERT (mBERT) model (Devlin et al., 2019) that supports 104 languages;
2. Cross-lingual language model (XLM-R)¹ (Lample and Conneau, 2019; Conneau et al., 2019) that supports 100 languages;
3. Language Agnostic SEntence Representations model (LASER)² (Artetxe and Schwenk, 2019) that supports 93 languages.

Among the three models, LASER is based on neural machine translation approach and strictly requires parallel data to train.

A common practice to train a large-scale multilingual model is to do so from scratch. *But do multilingual models always need to be trained from scratch? Can we transfer linguistic knowledge learned by English pretrained models to other languages?* In this work, we develop a technique to *rapidly* transfer an existing pretrained model from English to other languages. As the first step, we focus on building a *bilingual* language model (LM) of English and a target language. Starting from a pretrained English LM, we learn the target language specific parameters (*i.e.*, word embeddings), while keeping the encoder layers of the pretrained

¹<https://github.com/facebookresearch/XLM>

²<https://github.com/facebookresearch/LASER>